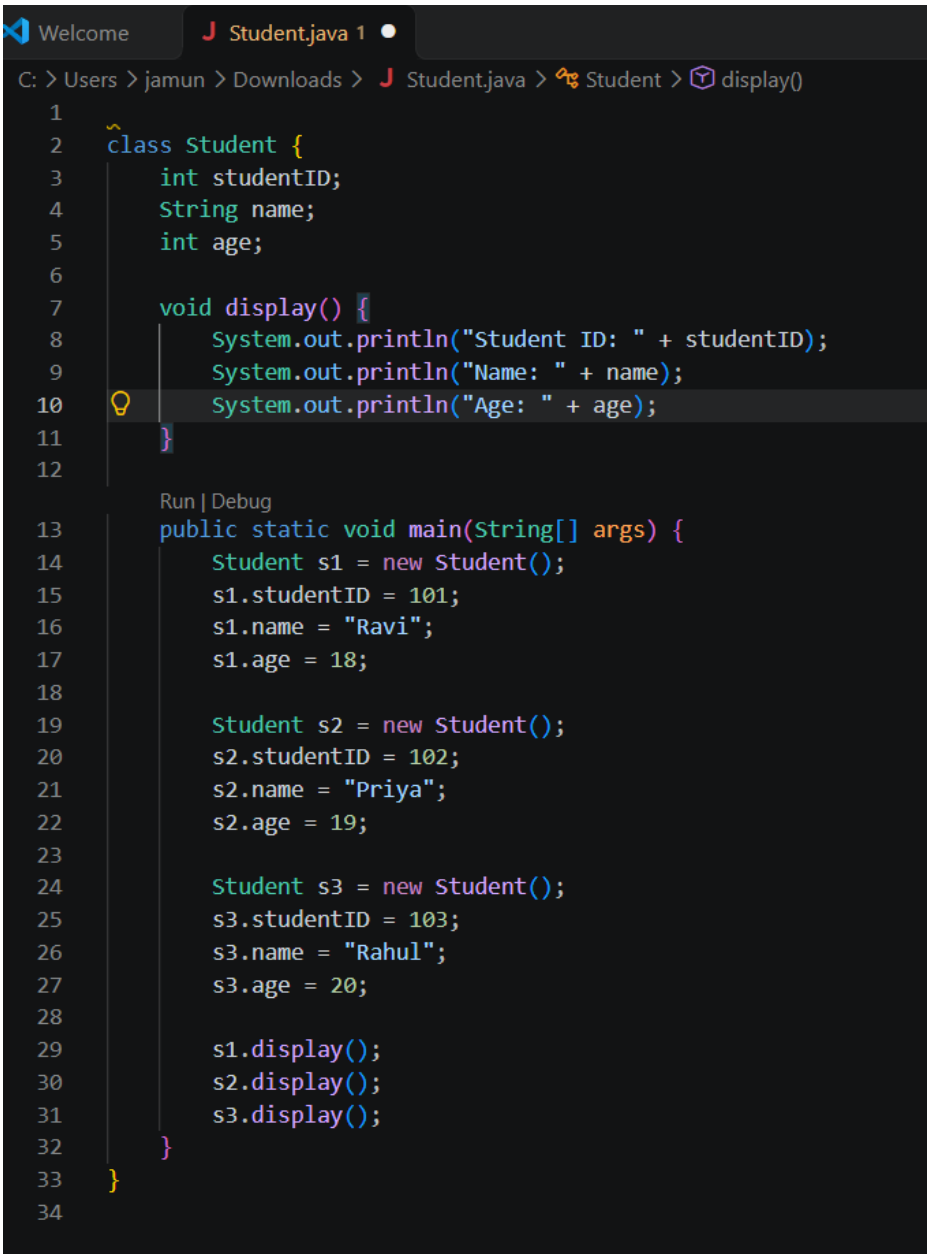
Week – 1

1. Create a class Student with student ID, name and age. Create three student objects and display their details.



Output-

```
PS C:\Users\jamun> cd Downloads
PS C:\Users\jamun\Downloads> javac Student.java
PS C:\Users\jamun\Downloads> java Student
Student ID: 101
Name: Ravi
Age: 18
Student ID: 102
Name: Priya
Age: 19
Student ID: 103
Name: Rahul
Age: 20
```

2. Create an Employee class with employee ID, name and salary. Initialize the values using a constructor and display employee details.

```java
class Employee {
    int employeeID;
    String name;
    double salary;

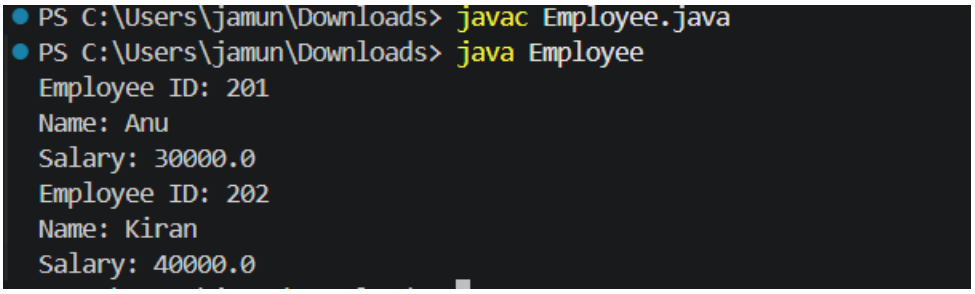
    Employee(int id, String n, double s) {
        employeeID = id;
        name = n;
        salary = s;
    }

    void display() {
        System.out.println("Employee ID: " + employeeID);
        System.out.println("Name: " + name);
        System.out.println("Salary: " + salary);
    }

    Run | Debug
    public static void main(String[] args) {
        Employee e1 = new Employee(id: 201, n: "Anu", s: 30000);
        Employee e2 = new Employee(id: 202, n: "Kiran", s: 40000)

        e1.display();
        e2.display();
    }
}
```

Output -



3. Create a BankAccount class with account number and balance. Implement deposit and withdrawal methods.

```java
class BankAccount {
    int accountNumber;
    double balance;

    BankAccount(int acc, double bal) {
        accountNumber = acc;
        balance = bal;
    }

    void deposit(double amount) {
        if (amount > 0) {
            balance = balance + amount;
            System.out.println("Deposit successful");
        }
    }

    void withdraw(double amount) {
        if (amount > 0 && amount <= balance) {
            balance = balance - amount;
            System.out.println("Withdrawal successful");
        } else {
            System.out.println("Insufficient balance or invalid amount");
        }
    }

    void display() {
        System.out.println("Account Number: " + accountNumber);
        System.out.println("Balance: " + balance);
    }

    Run | Debug
    public static void main(String[] args) {
        BankAccount b = new BankAccount(acc: 12345, bal: 5000);
        b.deposit(amount: 1000);
        b.withdraw(amount: 2000);
        b.display();
    }
}
```

Output -

```
PS C:\Users\jamun> cd Downloads
PS C:\Users\jamun\Downloads> javac BankAccount.java
PS C:\Users\jamun\Downloads> java BankAccount
Deposit successful
Withdrawal successful
Account Number: 12345
Balance: 4000.0
```

4. Create a Rectangle class with length and width. Calculate and display its area and perimeter.

```
class Rectangle {
    double length;
    double width;

    Rectangle(double l, double w) {
        length = l;
        width = w;
    }

    void calculate() {
        double area = length * width;
        double perimeter = 2 * (length + width);

        System.out.println("Area: " + area);
        System.out.println("Perimeter: " + perimeter);
    }

    Run | Debug
    public static void main(String[] args) {
        Rectangle r = new Rectangle(l: 10, w: 5);
        r.calculate();
    }
}
```

Output -

```
PS C:\Users\jamun> cd Downloads
PS C:\Users\jamun\Downloads> javac Rectangle.java
PS C:\Users\jamun\Downloads> java Rectangle
Area: 50.0
Perimeter: 30.0
```

5. Create a Book class with ISBN, title and price. Initialize the values using a parameterized constructor.

```java
class Book {
    String ISBN;
    String title;
    double price;

    Book(String i, String t, double p) {
        ISBN = i;
        title = t;
        price = p;
    }

    void display() {
        System.out.println("ISBN: " + ISBN);
        System.out.println("Title: " + title);
        System.out.println("Price: " + price);
    }

    public static void main(String[] args) {
        Book b = new Book("9781234567890", "Java Basics", 450.0);
        b.display();
    }
}
```

Output -

```
PS C:\Users\jamun\Downloads> javac Book.java
PS C:\Users\jamun\Downloads> java Book
ISBN: 9781234567890
Title: Java Basics
Price: 450.0
```